

PRACTICAL RECORD

BIOSIGNALS TO WELLBEING

(PSY-3027 / BIO-3414-1)

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Pulse Transducer

Aim: To record pulse rate using piezo-electric pulse transducer in the finger.

Principle: Piezo-electric material converts force applied into electricity. The electricity is used to record the pulse on the blood vessels that causes them to expand and contract.

Type of Device: Piezoelectric Sensor

Equipments: MatLab, Pulse Transducer, PowerLab 26T, Labchart 8

Method:

1. Connect the Pulse Transducer Input Cable on the Input socket of PowerLab 26T system
2. Switch on the PowerLab 26T Device
3. Start Labchart 8 Software and start "Scan Device" wait for the Green Mark on device
4. Select number of channels required to record the data from the software
5. Place the Pulse Transducer on participants middle finger on left hand
6. Select device input on the labchart software and setup the relevant parameters such as high pass filter, low pass filter, etc for the data before recording
7. Make conversation with the participant and ask them to stay calm for at least a minute.
8. Ask participant to not move their hand
9. Start data recording and record for the required time window
10. Slowly remove the electrode from the participant's finger
11. Save all data point in datapad and extract the data in .mat format
12. Run basic analysis and plotting with labchart

Discussion:

- Pulse transducers can provide insights into the functioning of the autonomic nervous system by tracking changes in heart rate in response to stress, relaxation, or emotional stimuli.
- They help detect irregularities in heart rhythms, such as bradycardia or tachycardia, and offer a non-invasive way to monitor pulse patterns over time.

Application:

- Pulse data can be used to study the physiological correlates of emotions, cognitive load, attention, and stress, enhancing our understanding of mind-body interactions.
- Pulse transducers are valuable in continuous patient monitoring, early detection of cardiovascular disorders, and evaluating the effectiveness of medications.

Electrocardiogram (ECG)

Aim: To record and analyze the electrical activity of the heart using an ECG sensor.

Principle: An ECG records the electrical impulses generated by the heart during depolarization and repolarization of the cardiac muscle, which correspond to different phases of the heartbeat.

Type of Device: Electrocardiogram

Equipments: MatLab, ECG Electrodes, PowerLab 26T, LabChart 8 Software

Method:

1. Attach 2 ECG electrodes to the participant below their right (red) and left (yellow) clavicle. Add the third electrode to their left arm (green).
2. Connect electrode cables to PowerLab 26T input and turn on the PowerLab device.
3. Start LabChart 8 and scan for the connected device.
4. Select required channels for ECG data.
5. Adjust filters and sampling rate.
6. Ask participant to stay still and relaxed.
7. Begin recording ECG for a specified time window.
8. Save data to the datapad and export in .mat format.
9. Run analysis to detect R-peaks and determine heart rate.

Discussion:

- ECG helps assess the electrical functioning of the heart, revealing heart rate, rhythm, and abnormalities like arrhythmias.
- It provides a non-invasive way to monitor the cardiac cycle and detect early signs of heart disease.

Application:

- Used in both research and clinical settings to monitor cardiac response to physical or psychological stress.
- Essential in diagnosing myocardial infarction, arrhythmias, and in evaluating cardiac medications.

Heart Rate Variability (HRV)

Aim: To measure heart rate variability to assess autonomic nervous system balance.

Principle: HRV measures the variation in time between successive R-peaks in the ECG signal. These variations are influenced by sympathetic and parasympathetic nervous system activity.

Type of Device: Electrocardiogram

Equipments: MatLab, PowerLab 26T, ECG electrodes, LabChart 8 Software

Method:

1. Set up ECG as described in the previous experiment.
2. Record ECG for 5–10 minutes or more depending on the aim of your study, under resting conditions.
3. Extract R-R interval data from the recorded ECG.
4. Use LabChart HRV module or MatLab to compute time-domain (e.g., SDNN, RMSSD) and frequency-domain (e.g., LF/HF ratio) parameters.
5. Export processed data in .mat format for further statistical analysis.

Discussion:

- HRV reflects the body's ability to adapt to stress and maintain homeostasis.
- A high HRV may indicate good autonomic flexibility, while low HRV is associated with stress, fatigue, or cardiovascular risk.

Application:

- Used in psychophysiological studies on stress, anxiety, and emotional regulation.
- Clinically, HRV is applied in cardiovascular risk assessment and biofeedback training.

Galvanic Skin Response (GSR)

Aim: To measure skin conductance changes using a GSR sensor.

Principle: GSR measures changes in skin conductance due to sweat gland activity, which is modulated by the sympathetic nervous system during emotional or physiological arousal.

Type of Device: Electrodermal Activity Sensor (GSR)

Equipments: MatLab, GSR Electrodes, PowerLab 26T, LabChart 8 Software

Method:

1. Record baseline GSR values by suspending the poles a few centimeters away from each other.
2. Attach GSR electrodes to the index and ring fingers of the participant's non-dominant hand.
3. Connect the input cable to PowerLab 26T, turn on the PowerLab device and start LabChart 8.
4. Select device input on the labchart software and setup the relevant parameters such as high pass filter, low pass filter etc for the data before recording.
5. Record baseline GSR for a few minutes, then expose participant to stimuli
6. Record the changes in skin conductance, save data and export in .mat format.
7. Analyze changes over time using LabChart or MatLab.

Discussion:

- GSR provides insight into emotional arousal and autonomic nervous system activation.
- It is often used as an index of psychological or physiological stress.

Application:

- Used in psychological experiments involving emotion, attention, or fear conditioning.
- Clinically and commercially applied in lie detection, stress monitoring, and human-computer interaction research.

Eye Tracking

Aim: To record eye tracking data using Tobii Pro eye tracker and software.

Principle: The eye tracker tracks the eye movement and gaze points. It emits infrared light from three sources and based on how the light is reflected back from the eye into the sensor, it determines the position and gaze of it.

Type of Device: Tobii Pro Fusion Eye Tracker

Equipments: Eye tracker, Tobii Pro software, Screen

Method:

1. Place the eye tracker at the base of the screen and connect them together.
2. Open the Tobii Pro software, create a new project and add stimuli. Make sure to place the configuration routine at the beginning of the series.
3. Give instructions to the participants and start the experiment.
4. Adjust the screen to ensure that the eyes are within the eye tracker's range and start configuration.
5. Post the experiment, go to the "analyze" tab and choose the participant name that you recorded the data under.
6. Now, you can view the playback of the stimuli and where the participants' gazes were falling on the screen.
7. You can also access gaze maps and heat maps for each stimulus. The numbers indicate the order in which the gazes followed each other and the size of the circle indicated the duration of the gaze.

Discussion:

- Eye tracking data provides valuable insights into visual attention and cognitive processing, allowing us to observe which elements of a visual stimulus draw attention.
- The use of gaze plots and heat maps helps in understanding patterns of visual exploration, such as whether the participant's attention is goal-directed or exploratory.

Application:

- In psychology and neuroscience, eye tracking is used to study attention, reading behavior, decision-making, and disorders like autism or ADHD.
- In marketing and human-computer interaction, it helps assess user experience, webpage design effectiveness, and how advertisements capture visual attention.

FLOWCHARTS

Pulse Transducer

Connect transducer to PowerLab

Open LabChart 8 and configure device setting

Place transducer on participant's finger

Start recording while participant remains still

Save and export data for analysis

GSR

Attach GSR electrodes to fingers

Connect device and configure in LabChart

Record baseline GSR readings

Expose participant to stimulus and record changes in skin conductance

Save, export, and analyze data

ECG

Attach ECG electrodes below right clavicle, left clavicle and left arm

Connect cables and turn on PowerLab

Start LabChart 8 and set filters

Record ECG while participant rests

Save, export data, and analyze R-peaks

HRV

Follow the steps above to record ECG data for 5-10 minutes. Extract and analyze R-R intervals from the ECG signal

Eye Tracker

Set up the eye tracker at the base of the screen and connect it.

Launch Tobii Pro, create a new project, add stimuli, and include the configuration routine at the start.

Give instructions to the participant and begin the experiment after ensuring their eyes are within range.

Record and complete the session, then go to the "Analyze" tab and select the participant's data.

Review results using playback, gaze maps, and heat maps showing gaze order and duration.